

Task 5 – The Discovery Phase (EDA)

Complete Implementation Guide

Purpose:

This guide is aligned with the PlaceMux Phase 1 Industry Immersion study guide. It explains every step required to complete the EDA task, why each step is performed, its business value, recommended dataset, candidate insights, follow-up questions, and how to maximize the evaluation score.

1. Task Requirements

- Deliver an EDA notebook with annotated charts.
- Prepare a shortlist of candidate insights.
- Prepare follow-up questions generated from the EDA.
- Use a real dataset and validate its quality.
- Demonstrate the notebook on real data.
- Handle missing values, duplicates, incorrect data types, and edge cases.
- Summarize findings using the 3–4 most decision-relevant charts.

2. Recommended Dataset

Recommended dataset: Customer Shopping Trends Dataset from Kaggle (3,900 rows, 18 columns). Use the original/raw CSV instead of a pre-cleaned version. Even if the dataset is already clean, perform validation checks and document the results. This demonstrates the complete analytical workflow.

3. Complete Workflow

Step 1: Import Libraries

What to do: Load pandas, numpy, matplotlib and seaborn.

Why this step is used: These libraries provide data manipulation, numerical operations and visualizations.

Step 2: Load Dataset

What to do: Read the CSV into a pandas DataFrame.

Why this step is used: Makes the dataset available for analysis.

Step 3: Preview Data

What to do: Use `head()`, `tail()`, `shape()`, `columns`, `info()`, `describe()`.

Why this step is used: Understand the structure, size, data types and summary statistics.

Step 4: Check Missing Values

What to do: Use `isnull().sum()`.

Why this step is used: Missing values can bias analysis and must be identified before EDA.

Step 5: Handle Missing Values

What to do: Fill or remove missing values if they exist.

Why this step is used: Improves data quality and prevents errors.

Step 6: Check Duplicates

What to do: Use `duplicated().sum()` and `drop_duplicates()`.

Why this step is used: Avoids counting the same transaction twice.

Step 7: Validate Data Types

What to do: Check dtypes.

Why this step is used: Ensures numeric columns are numeric and categorical columns are text.

Step 8: Check Unique Values

What to do: Review unique values in categorical columns.

Why this step is used: Detects spelling inconsistencies and invalid categories.

Step 9: Validate Invalid Values

What to do: Look for impossible values (negative age, negative purchase amount).

Why this step is used: Removes incorrect records.

Step 10: Detect Outliers

What to do: Use boxplots.

Why this step is used: Identifies unusually large or small values that may affect results.

Step 11: Univariate Analysis

What to do: Analyze one variable at a time using histograms, countplots and boxplots.

Why this step is used: Understand distributions, spread and frequencies.

Step 12: Bivariate Analysis

What to do: Compare two variables using bar charts, boxplots and scatter plots.

Why this step is used: Discover relationships between variables.

Step 13: Multivariate Analysis

What to do: Analyze three or more variables together.

Why this step is used: Identify deeper business patterns.

Step 14: Correlation Analysis

What to do: Create a correlation matrix/heatmap for numeric variables.

Why this step is used: Measures relationships between numerical features.

Step 15: Segmentation

What to do: Group data by Gender, Category, Season, Subscription Status, etc.

Why this step is used: Shows how customer behavior differs across groups.

Step 16: Identify Surprising Patterns

What to do: Investigate unexpected findings.

Why this step is used: Ensures insights are genuine and not data issues.

Step 17: Generate Candidate Insights

What to do: Summarize important discoveries.

Why this step is used: Converts analysis into business knowledge.

Step 18: Prepare Follow-up Questions

What to do: List questions raised by the EDA.

Why this step is used: Creates hypotheses for future analysis.

Step 19: Final Summary

What to do: Present the 3–4 most decision-relevant charts.

Why this step is used: Communicates findings clearly to stakeholders.

4. Candidate Insights

- Identify the most purchased product categories.
- Determine which customer segments spend the most.
- Compare purchase amounts across seasons.
- Evaluate whether subscription status is associated with higher spending.
- Identify the most common payment methods.
- Analyze review rating distribution.
- Compare spending across locations.
- Identify repeat purchase behavior.

5. Follow-up Questions

- Why do some customer groups spend more than others?
- Does using a promo code increase purchase amount?
- Which product categories generate the highest long-term value?
- Does shipping type influence customer ratings?
- Do repeat customers spend more than new customers?
- Which season should receive the highest marketing budget?
- What factors influence subscription status?
- Should marketing campaigns target specific age groups?